

Retail Sales Analysis

Turning Transaction Data into Business Insights

1. Executive Summary

This project analyzes historical retail transaction data to understand overall sales performance, customer and product contribution, regional differences, and sales patterns over time.

The analysis was performed using **Python, Pandas, NumPy, and Matplotlib** in Jupyter Notebook. The objective was to answer a set of business-focused questions and translate the results into clear findings that could support further investigation and decision-making.

The analysis found that **Technology was the highest-selling category**, while **West was the highest-performing region**. The **South region contributed approximately 17.2% of total sales**. Customer revenue was relatively distributed, with the **top 10 customers contributing approximately 6.8% of total sales**, while 168 customers accounted for 50% of cumulative sales.

The analysis also identified differences in monthly performance, with **November showing the strongest average monthly sales**, and **November 2018 standing out as an unusually strong month** that would require additional business data to understand its underlying cause.

2. Business Context

A retail business generates large amounts of transactional data across customers, products, categories, regions, and time periods.

Simply knowing total sales is not enough to understand business performance. Management may also need to know:

- Which categories generate the most sales?

- Which products contribute the most revenue?

- How concentrated is revenue among customers?

- Which regions perform strongly or weakly?

- How does sales performance change over time?

- Are there unusual periods that require investigation?

This project uses historical transaction data to answer these questions through exploratory and descriptive analysis.

3. Problem Statement

The business wants to understand its sales performance and identify important patterns across **products, categories, customers, regions, and time periods**.

The analysis focuses on:

1. Measuring overall sales and order performance.
2. Identifying high-performing categories and products.
3. Understanding customer sales concentration.
4. Comparing regional and geographic performance.
5. Examining monthly and yearly sales patterns.
6. Identifying unusual sales periods for further investigation.
7. Translating analytical results into practical business observations.

4. Dataset

The project uses a transactional retail sales dataset containing information about:

Orders
Customers
Customer segments
Regions
States
Categories
Sub-categories
Products
Order dates
Shipping dates
Sales

The dataset is analyzed at the transaction/line-item level, so a single order can contain multiple product records.

This distinction is important when calculating order-level metrics. For example, **Order ID** is used to identify unique orders rather than simply counting rows.

5. Analytical Approach

The analysis followed a business-question-driven approach.

Step 1 — Data Preparation

The dataset was loaded into Pandas and the date fields were prepared for time-based analysis.

Basic data exploration was performed to understand the structure and relevant fields before beginning the analysis.

Step 2 — Sales and Order Analysis

The analysis calculated:

- Total sales
- Unique orders
- Unique customers
- Average order value
- Median order value
- Mode order value

These metrics provide a basic view of the scale and distribution of sales activity.

Step 3 — Category and Product Analysis

Sales were aggregated across:

- Categories
- Sub-categories
- Individual products

This helped identify the main contributors to overall sales.

Step 4 — Customer Analysis

Customer-level sales were calculated to understand revenue concentration.

The analysis examined:

- Top customers
- Top 10 customer contribution
- Number of customers required to reach 50% of cumulative sales
- High-value customer distribution

Step 5 — Geographic Analysis

Sales were compared across:

- Regions

- States

- Region × category combinations

This helped identify geographic differences in sales performance.

Step 6 — Time-Based Analysis

Sales were examined across:

- Years

- Months

- Monthly averages

- Category-specific monthly performance

A simple **±2 standard deviation screening rule** was also used to flag unusually high or low monthly sales periods.

Importantly, an anomaly flag does not establish the reason for the unusual result.

6. Key Performance Indicators

KPI	Result
Total Sales	■2,261,536.78
Average Order Value	■459.48
Median Order Value	■151.88
Mode Order Value	■12.96
Top 10 Customer Sales Share	~6.8%
Top 10 Product Sales Share	~10.8%
Customers Accounting for 50% of Sales	168

These KPIs provide a high-level view of the sales base before moving into detailed product, customer, geographic, and time analysis.

7. Category Performance

Finding

Technology was the highest-selling category, generating approximately **36.59% of total sales**.

Category sales were:

Category	Sales	Share
Technology	■827,455.87	36.59%
Furniture	■728,658.58	32.22%
Office Supplies	■705,422.33	31.19%

Business Interpretation

Technology contributes the largest share of sales among the three categories.

However, the analysis only establishes the sales contribution. It does **not** establish why Technology performs better, because the dataset does not contain information such as marketing expenditure, pricing strategy, inventory levels, or customer preferences.

Business Question for Further Investigation

The business could investigate whether Technology's higher sales are associated with:

- Product availability

- Product pricing

Customer demand

Promotional activity

Product assortment

Additional business data would be required to answer these questions.

8. Product and Sub-Category Performance

The analysis identified **Phones** as the highest-selling sub-category.

The highest-selling individual product was:

Canon imageCLASS 2200 Advanced Copier — ■61,599.82

The top 10 products together contributed approximately **10.8% of total sales**.

Interpretation

Sales are not concentrated entirely in a small number of products. While the leading products make a meaningful contribution, the majority of sales comes from products outside the top 10.

This provides a useful basis for further product-level analysis, particularly around assortment and product availability.

9. Regional Performance

The analysis compared sales across the four regions.

Key Findings

West was the highest-performing region.

South was the lowest-performing region.

South generated approximately **■389,151.46**, representing about **17.21% of total sales**.

Business Interpretation

The difference between regional sales indicates that sales performance varies geographically.

However, lower sales in a region do not automatically mean that the region is underperforming operationally. Sales can be affected by factors that are not available in the dataset.

Therefore, the South region should be treated as an **area for further investigation**, rather than assuming a specific cause for its lower sales.

10. Customer Segment Analysis

The dataset includes customer segment classifications (Consumer, Corporate, Home Office).

Customers were analyzed by segment to understand whether different customer types contribute differently to sales.

Finding

Sales contributions vary by customer segment, indicating that different customer types have different purchasing patterns.

Business Interpretation

Understanding segment-level performance can help the business tailor its strategies, product offerings, and marketing approaches to each customer type.

Further Investigation

The business could analyze whether segment-specific performance is associated with product category, region, or order frequency.

11. Temporal Analysis

Sales were analyzed across years and months to identify changes in performance over time.

The analysis found that:

- Sales varied across years.

- Monthly performance was not uniform throughout the year.

- November had the strongest average monthly sales performance.**

- November 2018 showed unusually strong sales** and was flagged for further investigation.

Anomaly Analysis

A simple **± 2 standard deviation rule** was used as a screening method for unusual monthly sales.

The purpose was not to label the month as an error or determine its cause, but to identify periods that deserve additional investigation.

For example, understanding an unusually strong month would require information such as:

- Promotions

- Holiday campaigns

- Inventory availability

Large customer orders

Marketing campaigns

Those variables are not included in the current dataset.

12. Cross-Dimensional Analysis

The project also examined relationships between different business dimensions, including:

Region × Category

Segment × Category

Category × Month

High-value customers × Segment

High-value customers × Region

High-value customers × Category

These analyses provide more detailed context than looking at individual dimensions independently.

For example, a category may perform strongly overall while having different levels of contribution across regions or customer segments.

The purpose of this analysis was to identify **patterns worth investigating**, rather than claim causal relationships that cannot be established from the available transaction data.

13. Business Insights

1. Technology is the largest sales category

Technology generated **■827,455.87**, representing approximately **36.59% of total sales**.

What: Technology contributes the largest share of sales.

So what: It represents an important part of the current revenue mix.

Now what: Further analysis could examine the products, customers, pricing, and availability contributing to Technology sales.

2. Regional sales are uneven

West recorded the highest regional sales, while South generated the lowest share at approximately **17.21%**.

What: Sales contribution differs across regions.

So what: Regional performance is not evenly distributed.

Now what: Investigate the South region using additional customer, product, order, and operational data before determining the reason for the difference.

3. Revenue is not heavily concentrated in the top 10 customers

The top 10 customers contribute approximately **6.8% of total sales**.

What: The largest 10 customers represent a relatively small portion of total sales.

So what: Sales are spread across a wider customer base.

Now what: Analyze customer acquisition, retention, order frequency, and customer value to understand broader growth opportunities.

4. A small group of customers contributes a substantial portion of sales

Although the top 10 customers contribute only about 6.8%, **168 customers account for 50% of cumulative sales**.

What: A larger group of customers collectively contributes half of total sales.

So what: Customer concentration exists beyond the very top accounts.

Now what: Segment these customers by purchase frequency, value, region, and category to understand their contribution more deeply.

5. November deserves additional investigation

November showed the strongest average monthly sales performance, and November 2018 was identified as an unusually strong period.

What: November consistently stands out in the monthly analysis.

So what: There may be business factors associated with stronger November sales.

Now what: Compare November sales with promotion, holiday, inventory, campaign, and large-order information if those datasets become available.

14. Recommendations

The recommendations from this analysis are intended as **investigation priorities**, rather than claims about the causes of observed patterns.

Regional investigation

Investigate the South region to understand why its sales contribution is lower than other regions.

Category planning

Monitor Technology because it currently represents the largest category by sales.

Customer analysis

Continue analyzing the broader customer base rather than focusing only on the largest accounts, since sales are distributed across many customers.

Seasonal planning

Use the observed monthly patterns as a starting point for investigating category-specific demand and planning.

Anomaly investigation

Review unusual sales periods alongside promotions, holidays, inventory, and large-order information before drawing conclusions about their causes.

15. Limitations

This analysis has several important limitations.

No profit information

The dataset contains **Sales**, but does not contain Profit or Margin.

Therefore, the project evaluates **sales performance rather than profitability**.

A category with high sales cannot automatically be described as the most profitable category.

No discount information

The dataset does not provide sufficient discount information for analyzing the relationship between discounts and sales.

No marketing information

The analysis cannot determine whether campaigns or advertising caused changes in sales.

No inventory information

The analysis cannot determine whether product availability or stock-outs influenced sales.

No causal analysis

The project identifies patterns and relationships in historical sales data, but it does not establish causal relationships.

Anomaly detection is only a screening method

The ± 2 standard deviation approach is used to flag unusual monthly sales periods. It does not mean that an identified month is incorrect or that a specific business event caused the result.

16. Tools and Skills Demonstrated

Programming & Analysis

- Python
- Pandas
- NumPy

Visualization

- Matplotlib

Data Analysis

- Exploratory Data Analysis
- Data aggregation
- GroupBy analysis
- Descriptive statistics
- Customer analysis
- Product and category analysis
- Regional analysis
- Time-based analysis
- Basic anomaly screening

Business Analysis

- KPI calculation
- Business-question-driven analysis
- Pattern identification
- Business insight generation
- Communicating analytical findings

17. Final Conclusion

This project demonstrates how transactional retail data can be transformed into structured business insights using Python.

The analysis identified the strongest sales categories, products, customers, and regions, examined customer concentration, explored sales patterns over time, and highlighted unusual periods requiring further investigation.

The most important findings were that **Technology generated the largest category share, West was the highest-performing region, South contributed approximately 17.2% of total sales, and the top 10 customers contributed approximately 6.8% of total sales.**

Rather than assuming causes that cannot be supported by the available data, the analysis identifies areas where additional business information would be useful. This makes the project a practical example of **descriptive data analysis and business-question-driven problem solving.**

Project Outcome

- A complete Jupyter Notebook containing the analysis

- Visualizations supporting key findings

- The original retail dataset

- A management presentation summarizing the analysis

- Documentation of the analytical approach and findings

The project demonstrates the ability to move from **raw transactional data** → **analytical questions** → **quantitative findings** → **business interpretation.**